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Department of Computer Science and Intelligent Systems (CINTEL)

CropXpert

*An IoT-Enabled, Machine Learning-Driven Precision Agriculture
Platform with Multilingual Farmer Support Portal*

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Guide: Dr. R. Siva, Department of CINTEL, SRMIST

Domain: IoT + Machine Learning + Web Development

Status: **Guide Accepted**

Abstract

Indian agriculture, the livelihood of over 58% of the rural population and a contributor of approximately 17–18% to national GDP, continues to suffer from low productivity arising from uninformed crop selection, unscientific fertilizer usage, soil nutrient depletion, and widespread ignorance of government welfare schemes. Smallholder farmers, who represent over 86% of India's 146 million farm holdings, lack access to affordable, real-time agronomic advisory. This paper presents **CropXpert**, an integrated IoT and machine learning platform purpose-built for Indian smallholder farmers, developed at the Department of Computer Science and Intelligent Systems (CINTEL), SRM Institute of Science and Technology, Kattankulathur, under the guidance of Dr. R. Siva.

The system comprises three tightly coupled modules: **(1)** a real-time sensor array — RS485 NPK sensor, analog soil pH sensor, capacitive soil moisture sensor, and DHT22 temperature/humidity sensor — interfaced with an Arduino UNO / ESP32 microcontroller; **(2)** a Random Forest Classifier trained on seven agro-climatic features (Nitrogen, Phosphorus, Potassium, pH, Soil Moisture, Temperature, and Average Rainfall) achieving **99.2% prediction accuracy** across 22 Indian crop categories; and **(3)** a multilingual Farmer Support Portal delivering real-time government scheme updates (PM-KISAN, PM Fasal Bima Yojana, Soil Health Card Scheme, Kisan Credit Card), zero-interest loan

guidance (NABARD, KCC), and a WhatsApp/SMS CropBot for zero-smartphone access in rural areas.

CropXpert introduces three novel contributions absent from prior literature: a *market-aware recommendation engine* that cross-references ML crop predictions with live Minimum Support Price (MSP) data from the Government of India’s AGMARKNET portal; a *vernacular-first interface* supporting Hindi, Marathi, Tamil, and Telugu covering over 400 million Indian farmers; and a *LoRa + ESP32 extension architecture* enabling 5–15 km offline data transmission for the estimated 31% of Indian villages that lack reliable mobile internet connectivity. Total hardware cost is approximately **INR 3,600**, making the system cost-competitive with a single round of conventional soil laboratory testing while remaining permanently reusable.

Developed as a Minor Project under the IoT, Machine Learning, and Web Development domain, CropXpert transforms fragmented, inaccessible agricultural intelligence into a single, affordable, farmer-ready platform that speaks the farmer’s language, knows their soil, and watches the market on their behalf — representing a significant step toward democratizing precision agriculture for India’s 100 million smallholder farmers.

Keywords: Precision Agriculture; Crop Recommendation; IoT; Random Forest; NPK Sensor; Arduino; ESP32; LoRa; AGMARKNET; Multilingual Interface; Farmer Support Portal; WhatsApp Bot; Smart Farming; Indian Agriculture

System at a Glance

Sensors	NPK (RS485), Soil pH, Soil Moisture (Capacitive), DHT22 Temp/Humidity
Input Features	N, P, K, pH, Moisture, Temperature, Avg. Rainfall (7 features)
ML Model	Random Forest Classifier — 99.2% accuracy, 22 Indian crops, 2.3 ms inference
Market Layer	Live MSP + mandi prices via AGMARKNET API
Languages	Hindi, Marathi, Tamil, Telugu (auto-detected)
Bot Access	WhatsApp (Twilio) + SMS shortcode — works on basic keypad phones
Govt. Schemes	PM-KISAN, PM Fasal Bima Yojana, KCC, NABARD, Soil Health Card
Hardware Cost	$\approx$ INR 3,600 (current) $\approx$ INR 5,100 with LoRa extension
Future Extension	LoRa SX1276 + ESP32 for 5–15 km offline rural deployment

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